- Robot systems: Input from sensors, controls the motors
- Bar code scanners: Input/output for inventory control and shipping
- Automatic sprinklers: Used in farming to control the wetness of the soil

Medical

- Apnoea monitors: Detects breathing & alarms if the baby stops breathing
- Cardiac monitors: Measures heart functions
- Renal monitors: Measures kidney functions
- Drug delivery: Administers proper doses
- Cancer treatments: Controls doses of radiation, drugs, or heat
- Prosthetic devices: Increases mobility for the handicapped
- Dialysis machines: Performs functions normally done by the kidney
- Pacemaker: Helps the heart beat regularly

Failures involving embedded Systems

"A failure? So Embedded Technology is not God's greatest gift to humanity?" Well, slow down young inspired individual. Though they are few and are greatly dwarfed by the positives of this technology, it is always a good idea to know as much as you can about your playing field. While systems failures are generally complex and may not lie entirely in one engineering domain, there have been some instances where the problem has been traced back to a malfunction of the embedded system, be it hardware or a software issue. While these system failures are actually quite rare, it has happened and so deserves to be studied as there is always something to be learned from our mistakes. We have avoided the finger-pointing and biased case studies which you can find all over and managed to pick out some of the actual situations when embedded systems failed to work up-to expectations. Plus this issue is not an advertisement for Embedded Systems so we will show you the slightly weaker side of it as well. Have a look.

Patriot

On February 25, 1991, an American Patriot (Phased Array Tracking Intercept of Target) Missile in Dharan, Saudi Arabia, horribly failed to intercept an incoming Iraqi Scud Missile. This failure caused the death of 28 soldiers and 97 people were injured in the strike.

The problem here lay in the embedded system responsible for tracking calculations of the incoming missile. This computer keeps time as an integer value in tenths of a second, and stores it in register of 24 bits length. Later, to